Imran E Kibria

4^{th} YEAR PHD RESEARCHER IN SPEECH & AUDIO AI

Website | Github | LinkedIn | Google Scholar | kibria.5@osu.edu

Research Interests

Speech Quality Assessment, Speech Enhancement, Domain Generalization, Audio Foundation Models.

Education

Ohio State University — *Columbus, OH, USA*
Ph.D. in Computer Science & Engineering, GPA: 3.9 / 4.0 — *Aug 2023 - Present*
- Research: Neural Representations for Perceptual Speech Quality
- Advisor: Donald S. Williamson

National University of Sciences and Technology — *Islamabad, Pakistan*
BS in Electrical Engineering — *Sept 2016 - July 2020*
- Major: Signal Processing and Machine Learning
- Thesis: Real-time Adaptive Filtering for Robust Vibration Sensing

Experience

Graduate Teaching Associate — *Columbus, OH, USA*
Department of Computer Science & Engineering, Ohio State University — *Aug 2023 - Present*

Courses
- Software I: Software Components — Instructor (Su'25, Au'25, Au'26)
- Machine Learning and Statistical Pattern Recognition (Sp'26)
- Computer Programming in Python (Au'23, Sp'24, Sp'25)
- Introduction to Neural Networks (Au'24)

Graduate Research Associate — *Columbus, OH, USA*
Audio, Speech, & Perceptually-Inspired Research Lab, Ohio State University — *May 2023 - Aug 2023*

Design Engineer — *Islamabad, Pakistan*
Center for Advanced Research in Engineering — *Nov 2020 - May 2022*

Developer Intern — *Islamabad, Pakistan*
N-ovative Health Technologies — *July 2020 - Oct 2022*

Publications

Sharpness-Aware Minimization for Generalizable Speech Quality Assessment — *[In Review]*
Imran E Kibria, Donald S. Williamson
IEEE/ACM TASLP

A Generalization Strategy for Speech Quality Prediction: Domain-Specific to Unified Datasets — May 2026
Imran E Kibria, Ada Lamba, Donald S. Williamson
IEEE ICASSP 2026

AttentiveMOS: A Lightweight Attention-Only Model for Speech Quality Prediction — Aug 2025
Imran E Kibria, Donald S. Williamson
ISCA Interspeech 2025

Smartphone-Based Point-of-Care Urinalysis Assessment — July 2022
Imran E Kibria, Hussnain Ali, Shoab A. Khan
IEEE EMBC 2022

Select Projects

One Model for All Domains: Zero-Shot Speech Quality Assessment [Code]

- Analyzed factors influencing domain shift in MOS-label datasets.
- Improved generalization by optimizing model weights to a flat minima.
- Investigated loss landscape characteristics and quality of latent features.
- Reported improvement over AdamW optimizer on 8 of 12 evaluation sets.

Skills: Domain Generalization, Sharpness-Aware Minimization, TensorFlow, Pytorch, Python.

Lightweight AI for Perceptual Speech Quality on the Edge [Code]

- Designed MOS prediction network for edge use case using 86K params.
- Analyzed transformer variants & positional encodings as design choices.
- Countered subjective noise in MOS ratings using self-knowledge distillation.
- Wav2vec-level performance with 100× fewer parameters on SOMOS dataset.

Skills: Swin Transformers, Self-Knowledge Distillation, TensorFlow, Python, SHEET toolkit.

Self-Supervised Learning for Speech Quality Assessment [Code]

- Adapted AST, CLAP, HuBERT, wav2vec 2.0, and WavLM for MOS prediction.
- Evaluated transfer learning performance on the NISQA benchmark.
- Compared SSL representations for robustness and cross-dataset generalization.

Skills: Foundation Models, Transfer Learning, Domain Generalization, PyTorch, HuggingFace.

Smartphone-Based Urine Test Strip Analysis for At-Home Diagnostics [Code]

- Developed algorithm for automated urine test strip analysis using smartphone camera.
- Performed lighting & color calibration, then color matching in different colors spaces.
- Achieved 92% in-house accuracy; low-cost alternative to laboratory analyzers.

Skills: Image Preprocessing & Segmentation, Multivariate Regression, Color Spaces, Scikit-learn.

Technical Skills

Languages: Python, MATLAB, Java, C, C++, SQL, LaTex.

Deep Learning: Tensorflow, PyTorch, HuggingFace, Github

Speech Toolkits: SHEET, ESPNet, SpeechBrain

Awards & Honors

CCBS Summer Graduate Research Award — June 2026
Summer grant awarded by The Center for Cognitive and Brain Sciences (CCBS)

CSE Scarlet & Gray Award — Aug 2023
Merit-Based Scholarship, The Ohio State University (5-year award)

Best English Speaker — June 2017
SEECS (School of Electrical Engineering & Computer Science) Declamation Contest

Certificate of Excellence — Dec 2014
Third Place – Secondary Board Examination, Cadet College Hasanabdal